

Machine Learning (ML) in Hindi

Week-8 Assignment

Total points: 15

Topics covered:

- Kernel Support Vector Classifiers (K-SVC), Multi-class classification
- Support Vector Regressors (SVR)
- Naïve Bayes Classifiers
- Decision Trees

Instructions

- Each question is a 1-point multiple choice question (MCQ) with only one correct answer unless specified otherwise.
- Multiple select questions (MSQ) are 2-point questions with one or more than one answers correct
- For programming questions, use the default parameter settings unless specified otherwise.

1. [MCQ] How does SVM handle multi-class classification?

- a) It cannot be used for problems with more than two classes.
- b) By using strategies like 'one-vs-one' or 'one-vs-rest' to combine multiple binary classifiers.
- c) By creating a single, complex hyperplane that separates all classes at once.
- d) By converting the problem into a regression task

Answer: b) By using strategies like 'one-vs-one' or 'one-vs-rest' to combine multiple binary classifiers.

2. [MCQ] The Naïve Bayes classifier is called 'naïve' because of which key assumption?

- a) It can only be used for binary classification problems.
- b) It assumes that all features are equally important.
- c) It assumes that the training data has no missing values.
- d) It assumes that the features are conditionally independent given the class label.

Answer d) It assumes that the features are conditionally independent given the class label.

3. [MCQ] A data scientist is building a decision tree and finds that a particular node has a Gini impurity of 0. What does this signify?
- a) All samples at that node belong to a single class.
 - b) The node is the root node of the tree.
 - c) The feature used to split this node was a continuous variable.
 - d) The feature used to split this node was a categorical variable.

Answer a) All samples at that node belong to a single class.

4. [MCQ] Which decision tree algorithm can handle both classification and regression tasks and typically creates binary splits?
- a) C4.5
 - b) CART
 - c) Naïve Bayes
 - d) ID3

Answer b) CART

5. [MCQ] In the one-vs-one (OvO) approach for multiclass SVM, how many binary classifier models are generated for N different classes?
- a) N
 - b) N-1
 - c) $N*(N-1)/2$
 - d) $N*(N+1)/2$

Answer c) $N*(N-1)/2$

6. [MSQ] What are the limitations of the Naïve Bayes classifier?
- a) Handles high-dimensional data well.
 - b) Sensitivity to feature interactions.
 - c) Robust to irrelevant features.
 - d) The independence assumption may not hold true in all datasets.

Answer b), d)

7. [MCQ] In a Regression Tree, what is the most common metric used to decide on the best split at a node?
- a) Information Gain
 - b) Misclassification Error
 - c) Gini Impurity
 - d) Mean Squared Error (MSE)

Answer d) Mean Squared Error (MSE)

8. [MCQ] In the context of Bayes' Theorem, $P(y|x) = \frac{P(x|y)P(y)}{P(x)}$ what does the term $P(y)$ represent?
- a) The Evidence
 - b) The Prior probability
 - c) The Posterior probability
 - d) The Likelihood

Answer b) The prior probability

9. [MCQ] A generative model like Naïve Bayes can be used to generate new synthetic samples, while a discriminative model like SVM cannot. Why?
- a) Because Naïve Bayes learns the underlying data distribution $P(x | y)$, which can be sampled from.
 - b) Because SVMs can only be used for binary classification.
 - c) Because SVMs are too computationally complex.
 - d) Because Naïve Bayes assumes feature independence.

Answer a) Because Naïve Bayes learns the underlying data distribution $P(x | y)$, which can be sampled from.

10. [MCQ] In a decision tree, if a node contains 10 red dots and 0 blue dots, it is considered:
- a) A root node.
 - b) An impure node.
 - c) A pure node.
 - d) An invalid node.

Answer c) A pure node.

11. [MCQ] The main purpose of a splitting criterion like Gini Impurity or Entropy is to
- a) ensure that the same number of samples go down each branch.
 - b) choose a feature that makes the resulting groups as pure as possible.
 - c) use every feature in the dataset at least once.
 - d) make the tree as deep as possible.

Answer b) choose a feature that makes the resulting groups as pure as possible.

12. [MSQ] Which of the following are applications of the Naive Bayes classifier?
- a) Email Spam Filtering
 - b) Sentiment Analysis

- c) Recommendation System
- d) Fraud Detection

Answer a), b), c), d)

13. [MCQ] Which of the following techniques CANNOT help mitigate overfitting in decision trees?

- a) Post-pruning the tree using a validation set.
- b) Setting a minimum number of samples required to split a node.
- c) Using Gain Ratio instead of Information Gain as the splitting criterion.
- d) None of the above

Answer c) Using Gain Ratio instead of Information Gain as the splitting criterion.